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Abstract

AgroPrice SL is a Digital Public Good designed to eliminate agricultural market price inequality in Sierra Leone. Over 60% of Sierra Leone's population depends on farming, yet smallholder farmers consistently receive below-market prices for their produce due to a critical lack of price information. Middlemen exploit this gap, deepening rural poverty.

This report presents a complete software design for AgroPrice SL — a web-based and SMS-accessible Farmer Market Price Information System. The system delivers real-time commodity prices, SMS alerts, market maps, and analytics, built on a three-tier architecture (React.js, Node.js, PostgreSQL) using the Agile SDLC. The project includes UML diagrams, database schema, a 10-screen Figma prototype, GitHub repository, and project management tools.

AgroPrice SL aligns directly with UN SDG 1 (No Poverty) and is licensed under the MIT License as an open-source public good.

Keywords: Digital Public Good, AgroPrice SL, Agile SDLC, Sierra Leone, Farmer Market Prices, SDG 1, SMS Alerts, PostgreSQL.

Chapter 1: Introduction

1.1 Background

Sierra Leone is an agrarian nation where over 60% of the population depends on agriculture. Despite this, smallholder farmers remain among the poorest citizens due to a critical lack of market price information. When farmers arrive at markets, they have no knowledge of prevailing commodity prices, allowing traders to offer prices far below market value. This information asymmetry perpetuates rural poverty.

1.2 Problem Statement

- Farmers sell crops at below-market prices due to price ignorance.
- Middlemen exploit the information gap to dictate unfair prices.
- Post-harvest losses occur because farmers cannot identify optimal selling times.
- Women farmers and rural communities are disproportionately affected.

1.3 Project Overview

AgroPrice SL is a Digital Public Good — a web-based and SMS-accessible Farmer Market Price Information System that provides real-time agricultural commodity prices across Sierra Leone's districts and markets.

1.4 Objectives

- Provide real-time, district-specific agricultural commodity prices.
- Deliver SMS price alerts for farmers using basic mobile phones.
- Enable market officers to enter and update price data easily.
- Display price trends and market locations on interactive maps.
- Operate as an open-source Digital Public Good under the MIT License.

1.5 SDG Alignment

AgroPrice SL directly aligns with UN Sustainable Development Goal 1: No Poverty — empowering farmers to earn fair value for their harvest, reduce exploitation, and improve livelihoods across Sierra Leone.

Chapter 2: Software Development Life Cycle (SDLC)

2.1 Selected Model: Agile

The Agile SDLC was selected for AgroPrice SL because requirements may evolve as farmer feedback is gathered, sprints enable iterative delivery of working features, and the model supports stakeholder collaboration from farmers to policymakers. Its flexibility also accommodates Sierra Leone's unique infrastructure constraints, including low bandwidth and limited smartphone access.

2.2 Functional Requirements

ID	Requirement
FR01	Farmers can view current market prices by commodity and district
FR02	System sends SMS price alerts to subscribed farmers
FR03	Admin can add, update, and delete commodity price data
FR04	Users can register and log in securely using JWT
FR05	System displays price trends and historical data charts
FR06	System works on low-bandwidth connections and feature phones

2.3 SDLC Phases Summary

Phase	Duration	Key Deliverable
1. Planning	3 weeks	Project Charter, Risk Register
2. Requirements	2 weeks	Functional & Non-Functional Specs
3. System Design	3 weeks	Architecture, DB Schema, UML
4. Implementation	9 weeks	React.js Frontend, Node.js Backend
5. Testing	3 weeks	UAT, Security, Performance Tests
6. Deployment	1 week	Live on agroprice.sl
7. Maintenance	Ongoing	Updates, Monitoring, Support

Chapter 3: System Architecture Design

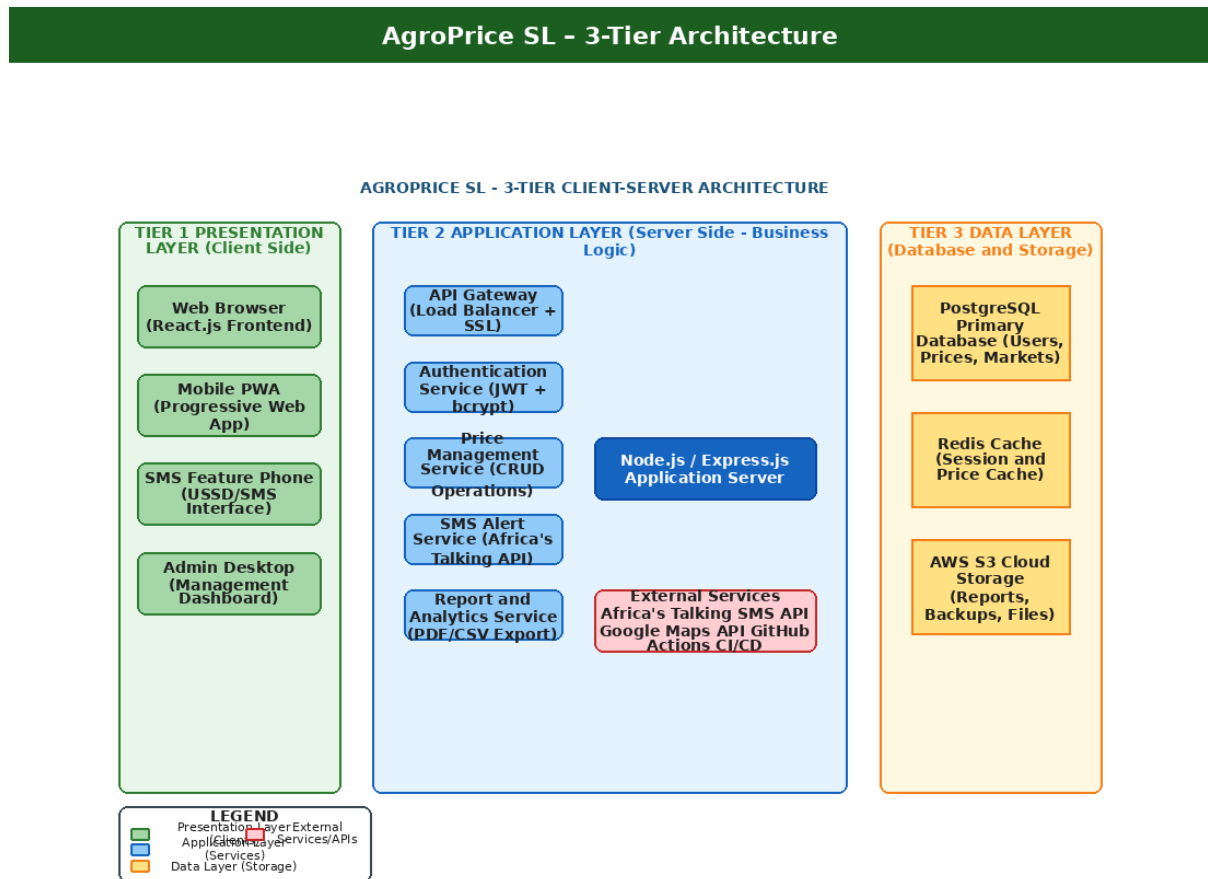
3.1 Architecture Overview

AgroPrice SL uses a Three-Tier Architecture: Presentation Layer (React.js web, SMS, PWA), Application Layer (Node.js/Express.js API services), and Data Layer (PostgreSQL, Redis, AWS S3). This ensures scalability, security, and separation of concerns.

Layer	Components
Presentation	Web Browser (React.js), Mobile PWA, SMS Feature Phone Interface
Application	API Gateway, JWT Auth Service, Price Service, SMS Alert Service, Analytics
Data	PostgreSQL Database, Redis Cache, AWS S3 Storage
External	Africa's Talking SMS API, Google Maps API

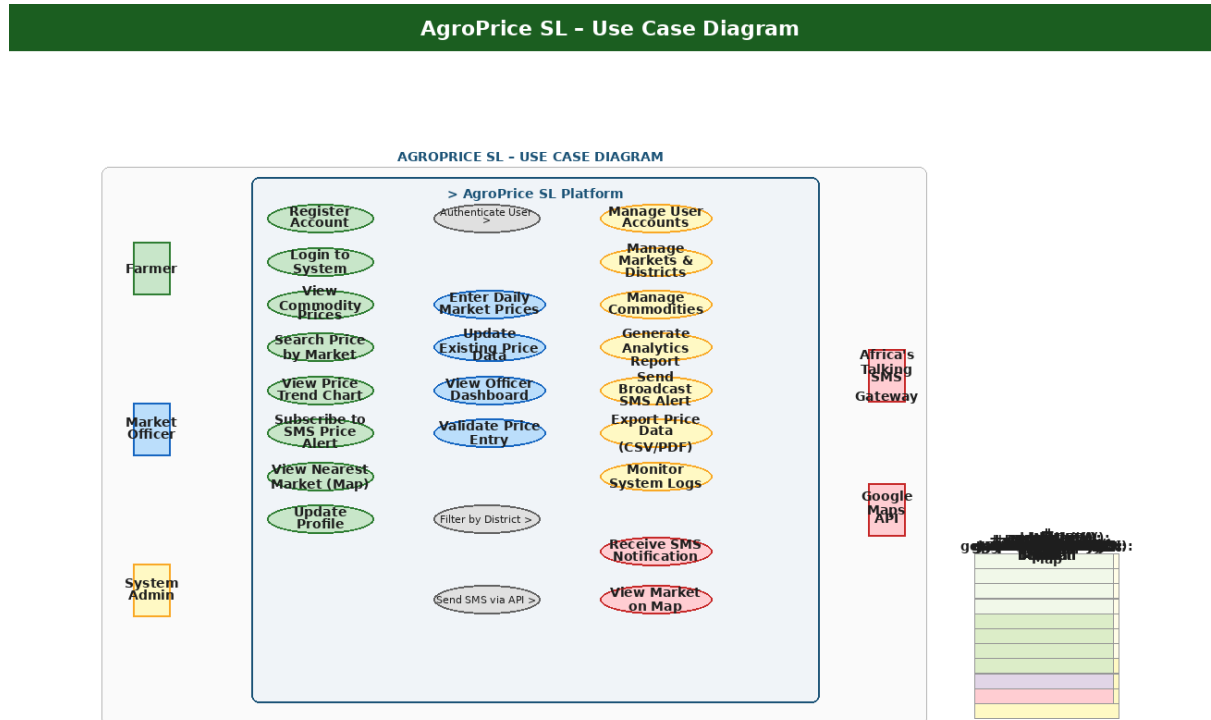
3.2 Architecture Diagram

Figure 1: AgroPrice SL – 3-Tier System Architecture



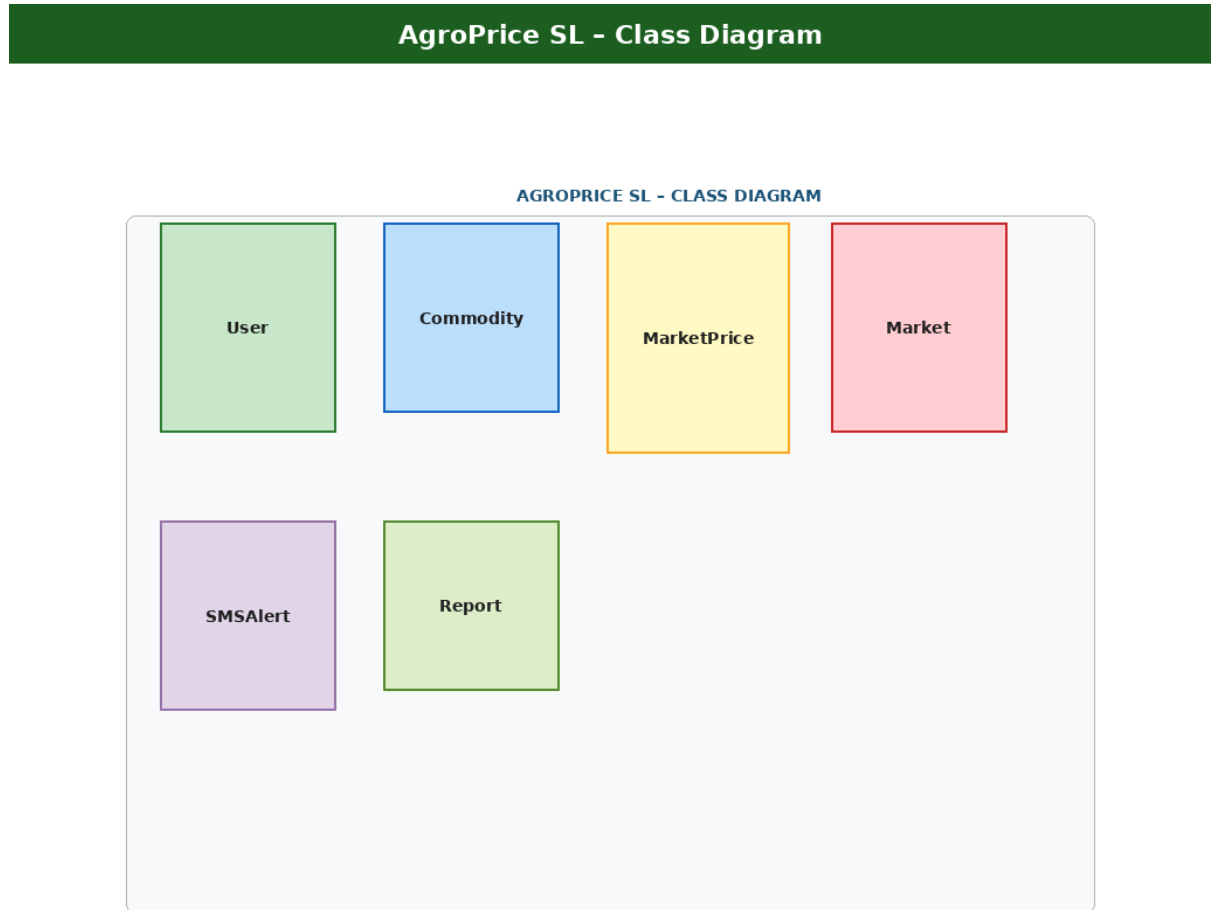
3.3 Use Case Diagram

Figure 2: Use Case Diagram – showing all actors (Farmer, Market Officer, Admin) and their system interactions.



3.4 Class Diagram

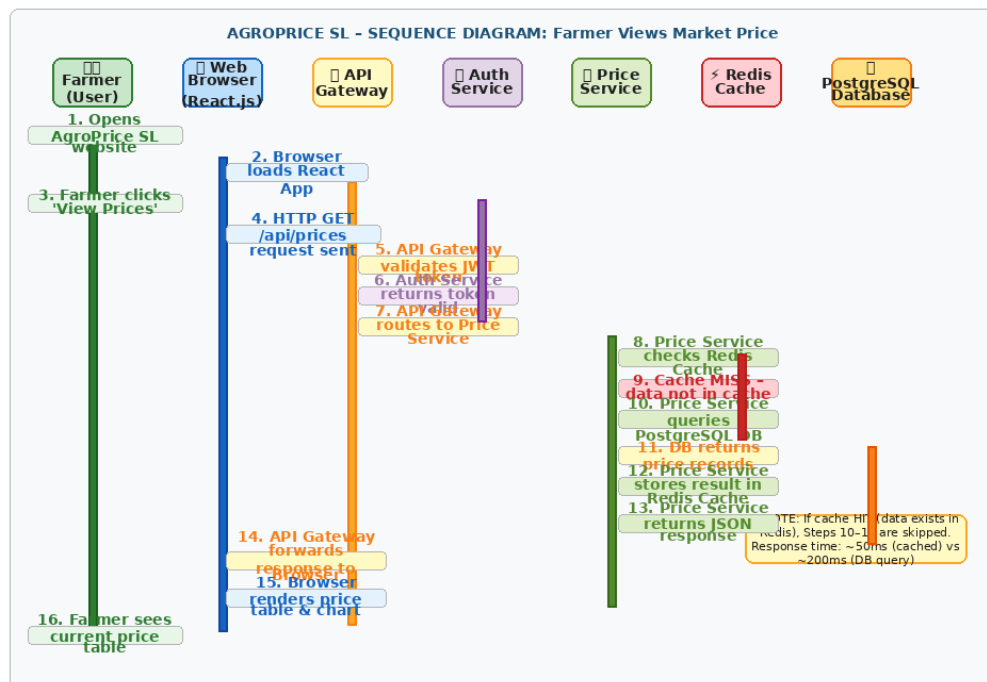
Figure 3: Class Diagram – showing core classes: User, Farmer, Commodity, Market, MarketPrice, SMSAlert, Report.



3.5 Sequence Diagram

Figure 4: Sequence Diagram – illustrating the price-viewing flow from Farmer → API → Database → Response.

AgroPrice SL - Sequence Diagram



Chapter 4: Database Design

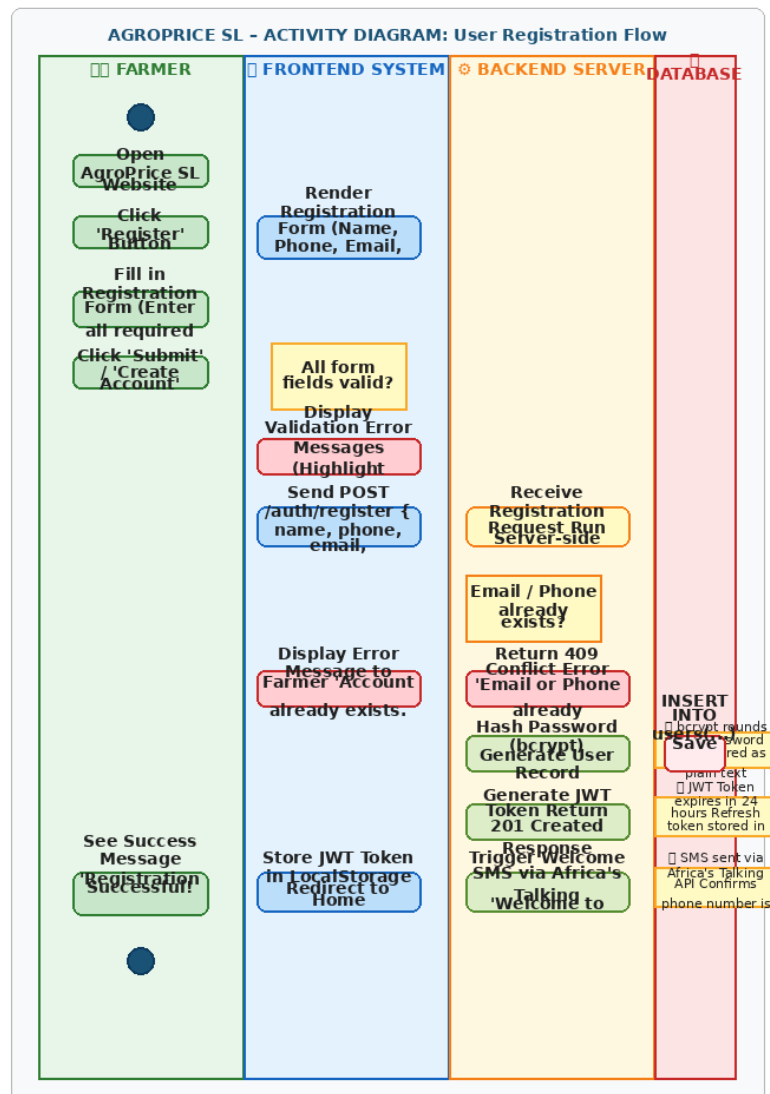
4.1 Overview

AgroPrice SL uses PostgreSQL 15 normalized to 3NF. The database has five core tables: USERS, COMMODITIES, MARKETS, MARKET_PRICES, and SMS_ALERTS.

4.2 Database ERD

Figure 5: Entity Relationship Diagram – showing all 5 tables with PK/FK relationships.

AgroPrice SL - Activity Diagram



4.3 Key Table Schemas

USERS Table

Column	Type	Constraint
user_id	INT	PRIMARY KEY, AUTO_INCREMENT
full_name	VARCHAR(100)	NOT NULL
phone_number	VARCHAR(20)	UNIQUE, NOT NULL
role	ENUM	farmer admin officer policymaker
password_hash	VARCHAR(255)	NOT NULL
district_id	INT	FOREIGN KEY

MARKET_PRICES Table (Core)

Column	Type	Constraint
price_id	INT	PRIMARY KEY, AUTO_INCREMENT
commodity_id	INT	FOREIGN KEY → COMMODITIES
market_id	INT	FOREIGN KEY → MARKETS
price_per_unit	DECIMAL(10,2)	NOT NULL
recorded_date	DATE	NOT NULL
trend	ENUM	rising falling stable
recorded_by	INT	FOREIGN KEY → USERS

Chapter 5: Prototype Development

5.1 Design Overview

The AgroPrice SL UI was designed in Figma following mobile-first, accessibility-focused principles. The design targets farmers with varying digital literacy levels, using large fonts, icon-based navigation, and a green agricultural colour palette. All 10 screens meet WCAG 2.1 AA accessibility standards.

5.2 Figma Link

[https://www.figma.com/\[insert-your-prototype-link-here\]](https://www.figma.com/[insert-your-prototype-link-here])

5.3 Mobile Screens (Phone Prototype)

Figure 6: Figma Mobile Screens – Screens 1 to 5 (Splash, Login, Registration, Dashboard, Market Selection)

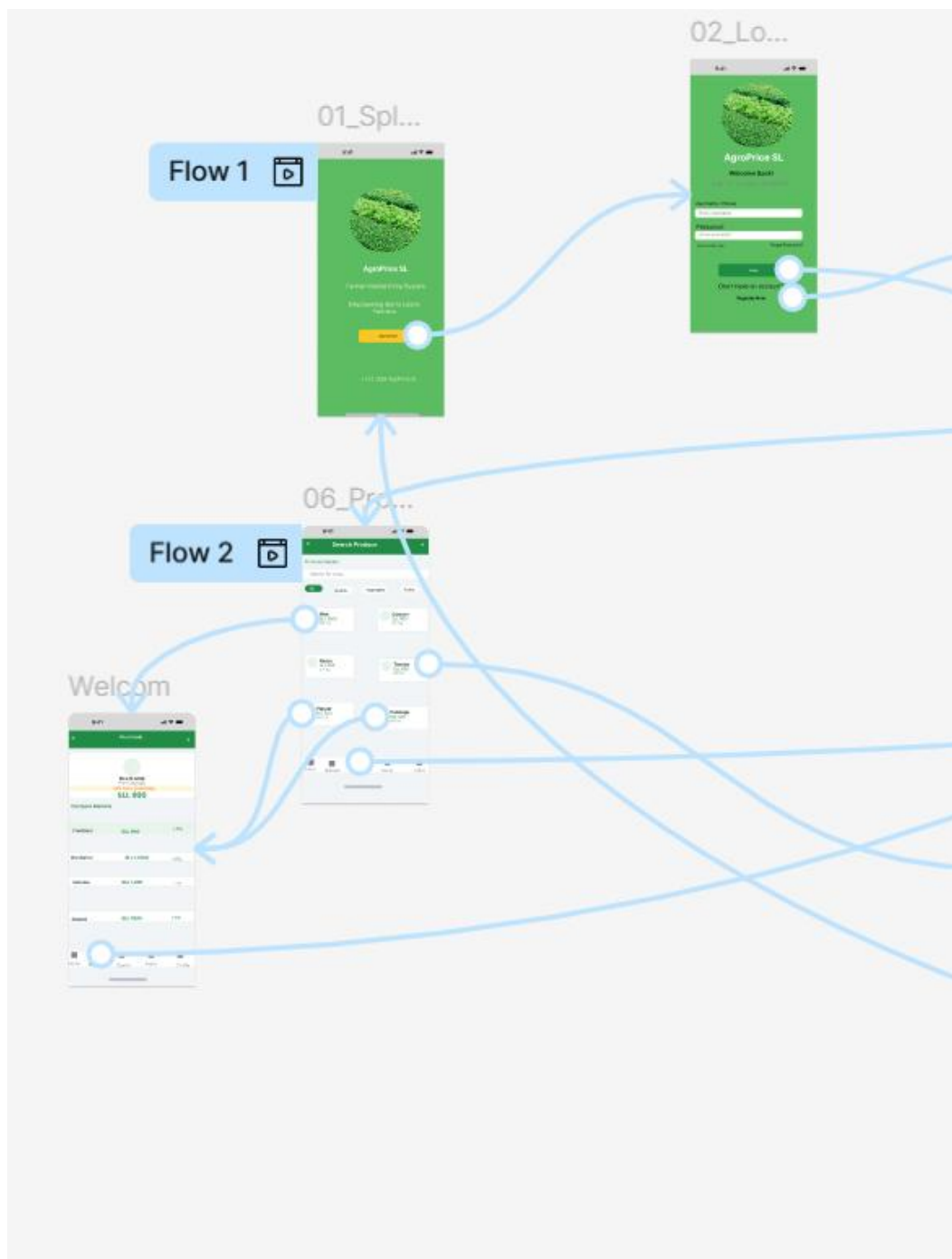
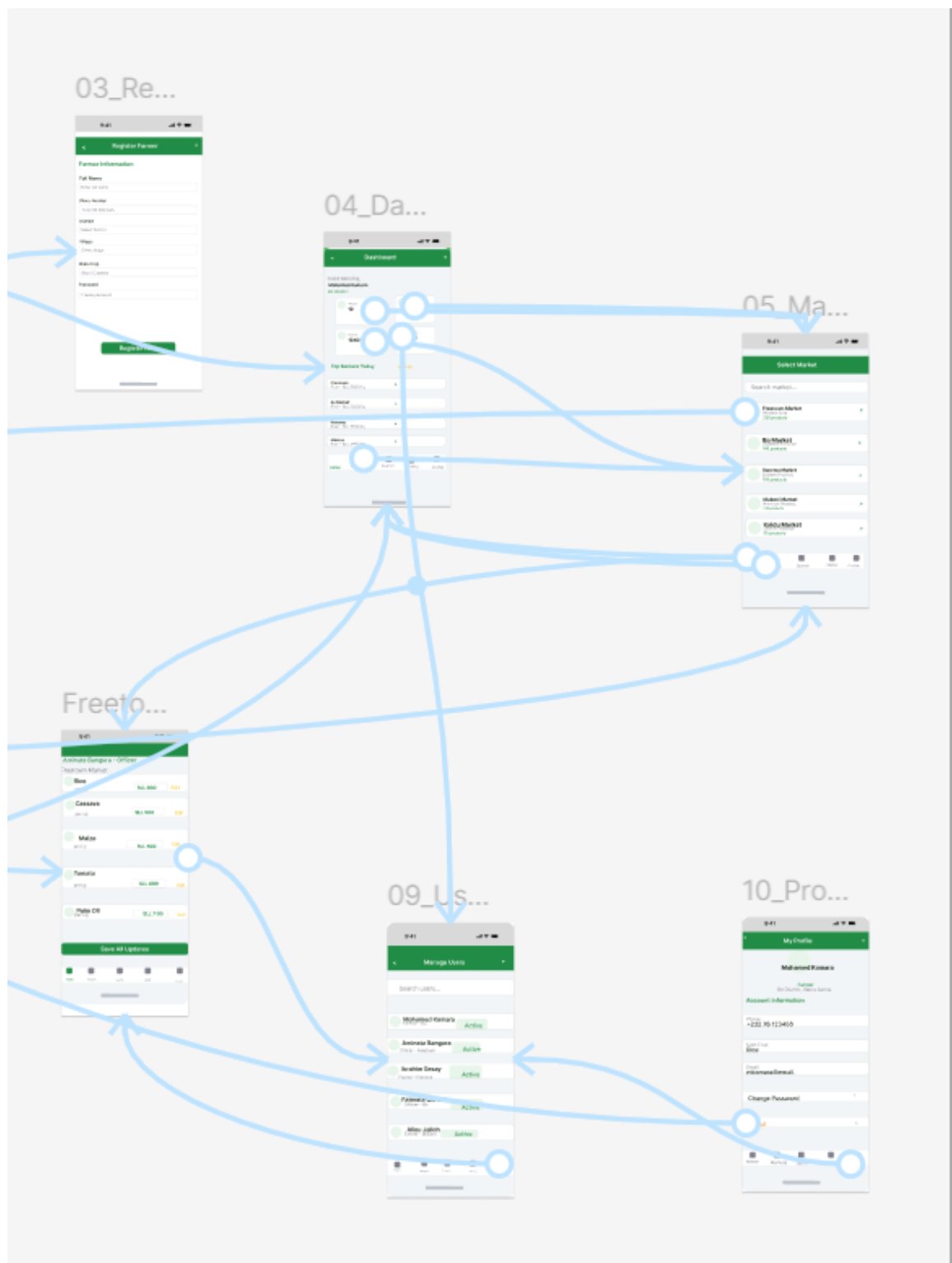


Figure 7: Figma Mobile Screens – Screens 6 to 10 (Produce Search, Price List, User Management, Profile)



5.4 Desktop Screens

Figure 8: Figma Desktop Screens – Screens 1 to 5 (Welcome, Login, Registration, Dashboard, Market Selection)

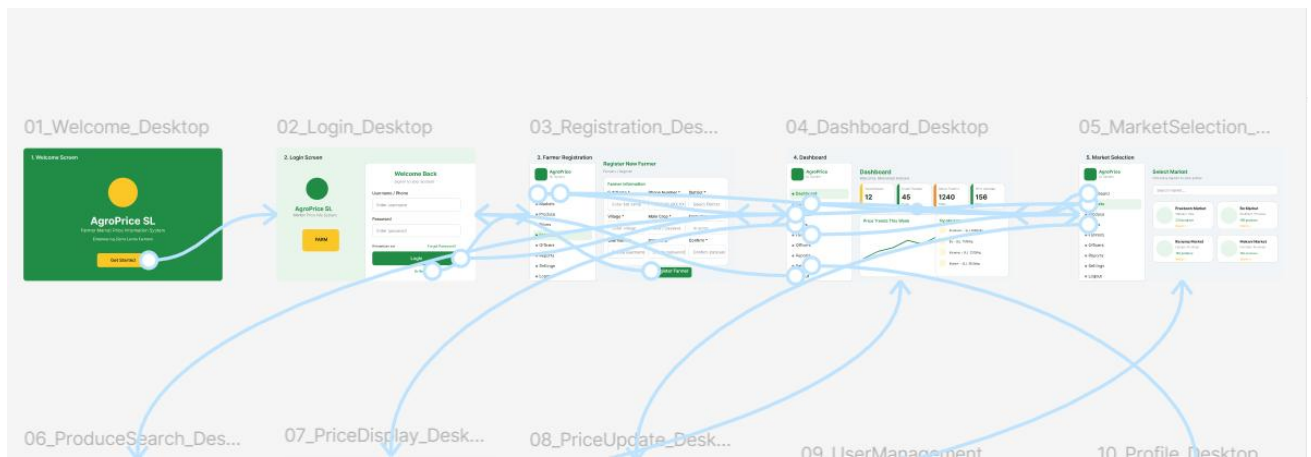
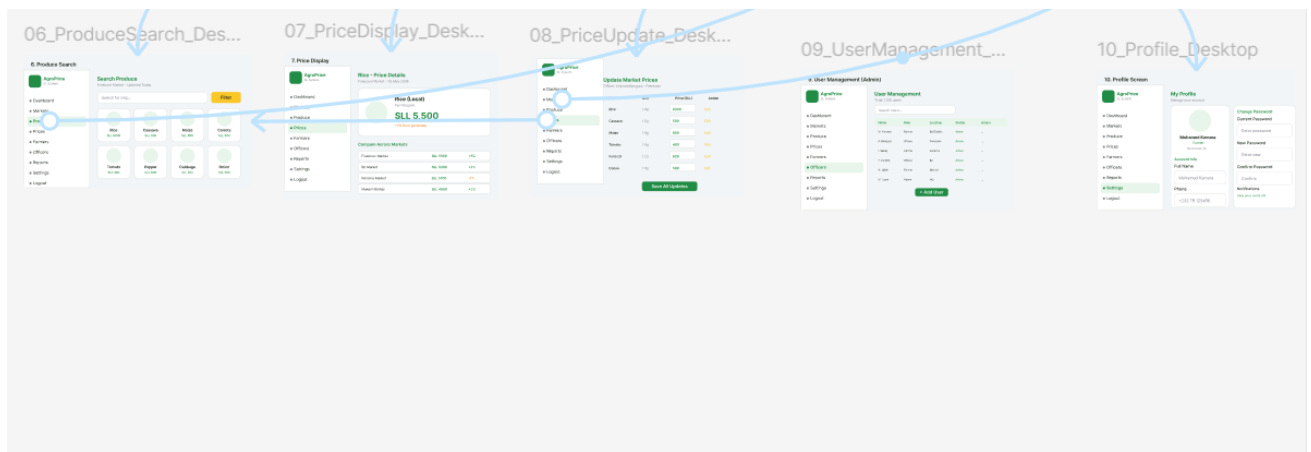


Figure 9: Figma Desktop Screens – Screens 6 to 10 (Produce Search, Price Display, Price Update, User Management, Profile)



Chapter 6: Git & GitHub Integration

6.1 Repository Overview

Property	Detail
Repository Name	agroprice-sl
GitHub URL	https://github.com/[yourusername]/agroprice-sl
License	MIT License
Visibility	Public
Default Branch	Main

6.2 Repository Structure

agroprice-sl/ |— docs/ (report, diagrams, ERD) |— frontend/ (React.js src, components, pages) |— backend/ (Node.js, Express, controllers, routes) |— database/ (schema.sql) |— prototype/ (Figma exports) |— README.md |— LICENSE

6.3 Branch Strategy

- main — Production-ready stable code only
- develop — Integration branch for completed features
- feature/* — Individual branches per feature (e.g. feature/user-auth)
- hotfix/* — Emergency fix branches from main

6.4 GitHub Screenshot

Figure 10: Insert screenshot of your GitHub repository page here.

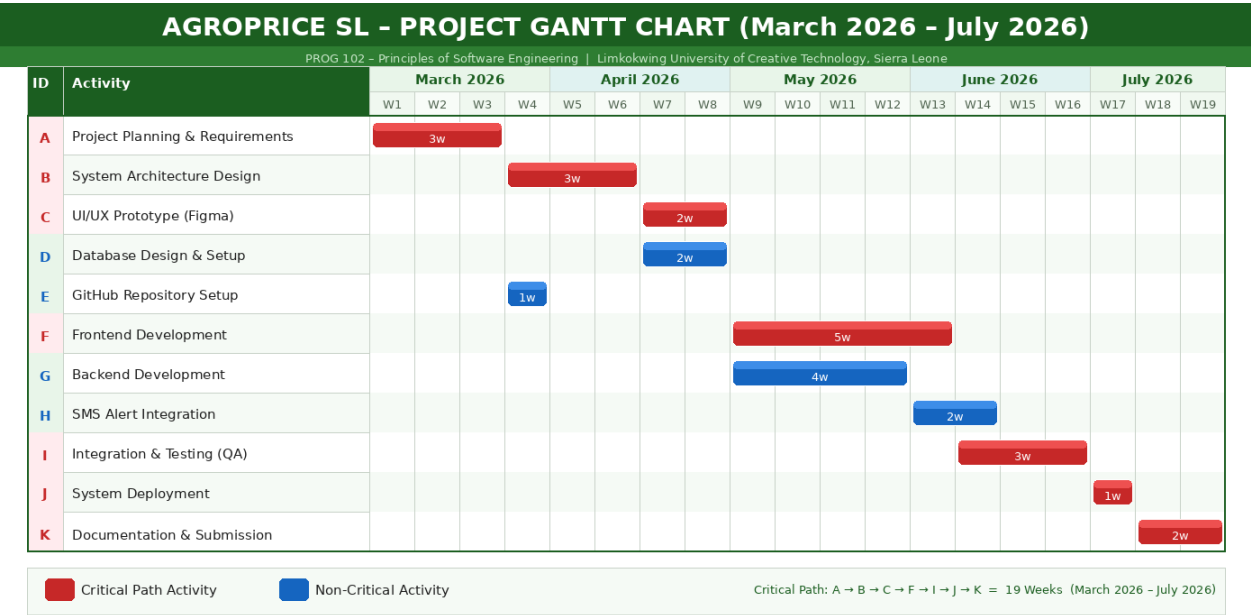
Chapter 7: Project Management

7.1 Project Activities

ID	Activity	Duration	Depends On
A	Project Planning & Requirements	3 wks	–
B	System Architecture Design	3 wks	A
C	UI/UX Prototype (Figma)	2 wks	B
D	Database Design & Setup	2 wks	B
E	GitHub Repository Setup	1 wk	A
F	Frontend Development	5 wks	C
G	Backend Development	4 wks	D
H	SMS Alert Integration	2 wks	G
I	Integration & Testing	3 wks	F,G
J	Deployment	1 wk	I
K	Documentation & Submission	2 wks	J

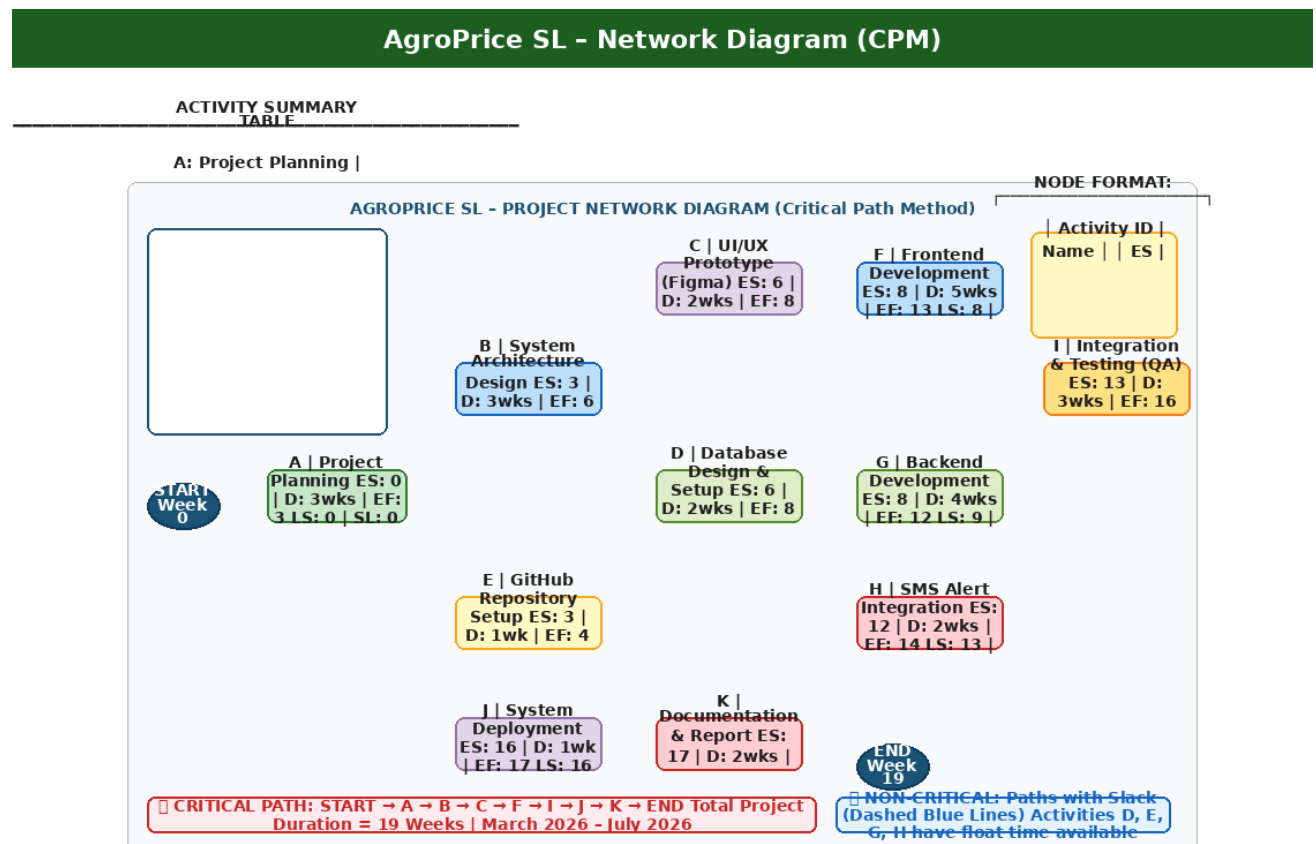
Critical Path: START → A → B → C → F → I → J → K → END (Total: 19 Weeks)

7.2 Gantt Chart



7.3 Network Diagram (CPM)

Figure 12: Network Diagram – Critical Path Method showing activity sequence and slack.



Chapter 8: License, Design Quality and SDG Relevance

8.1 Software License – MIT

AgroPrice SL is released under the MIT License, making it a true Digital Public Good. This permissive license allows governments, NGOs, and development partners to freely use, modify, redistribute, and deploy the system. It aligns with Digital Public Goods Alliance (DPGA) open licensing standards.

Resource	License	Usage
React.js	MIT	Frontend UI Framework
Node.js	MIT	Backend Runtime
PostgreSQL	PostgreSQL (Open Source)	Primary Database
Africa's Talking	Commercial SaaS	SMS Gateway
Chart.js	MIT	Analytics Charts
Bootstrap 5	MIT	Responsive Design

8.2 UI/UX Design Quality

- **Simplicity:** Maximum 3 clicks to reach any core feature.
- **Accessibility:** WCAG 2.1 AA compliant, minimum 4.5:1 colour contrast.
- **Responsiveness:** Mobile-first design from 320px screen width.
- **Agricultural Identity:** Green palette builds trust and brand recognition.
- **Typography:** Poppins Bold (headings), Poppins Regular (body), Roboto Mono (prices).

8.3 SDG 1 Alignment Matrix

Problem in Sierra Leone	AgroPrice SL Solution	Impact
Farmers sell below market due to price ignorance	Real-time prices via web and SMS	15–25% income increase
Middlemen exploit information gap	Transparent public price publishing	Fairer market transactions
Post-harvest losses from poor timing	Price trend analytics	Better selling decisions
Women farmers lack market access	SMS access via feature phones	Inclusive financial access

9: Conclusion and Recommendations

9.1 Summary

AgroPrice SL delivers a complete software design addressing market information inequality in Sierra Leone's agricultural sector. This project has successfully produced all seven required deliverables: SDLC documentation, system architecture, UML diagrams, database schema, Figma prototype, GitHub integration, and project management tools — all aligned to SDG 1: No Poverty.

9.3 Recommendations

- Integrate AI price prediction using historical commodity data.
- Add Krio, Mende, and Temne language support for broader reach.
- Seek Ministry of Agriculture endorsement and funding for district officers.
- Expand to Android app for improved offline capability.

9.4 Conclusion

AgroPrice SL is a practical, scalable, and impactful response to one of Sierra Leone's most persistent development challenges. Every farmer who receives an SMS price alert and uses it to negotiate a fair price moves one step further from poverty. This system is designed to make that happen — at scale, across Sierra Leone.

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